Fall 2025 CPTS530 Homework 03 Report
Numerical Analysis

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Due: Tuesday, September 30, 2025

Contents

Question 1

Problem Statement: Suppose that $x > 0$, $y < 0$. Using definition of absolute value and splitting into cases, show that $|x + y| \leq |x - y|$.

Solution:
Given: $x > 0$, $y < 0$
To show: $|x + y| \leq |x - y|$

Visual Representation:

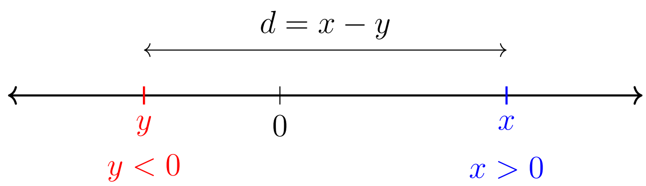

Figure 1: Number line showing the relationship between x, y and $x - y$

Since $x > 0$ and $y < 0$, we can assume that $x - y = d$ where $d > 0$.
Therefore, RHS $= |x - y| = d$ is always true.
Now we need to consider cases for $|x + y|$ based on whether $x + y$ is positive or negative.

Case I: $x + y \geq 0$
From $x - y = d$, we get $x = y + d$.
Therefore: $x + y = (y + d) + y = 2y + d$
Since $x + y \geq 0$, we have $|x + y| = x + y = 2y + d$.
So LHS $= 2y + d$ and RHS $= d$.
Since $y < 0$, we have $2y < 0$, which means LHS $= 2y + d < d =$ RHS.

Case II: $x + y < 0$
From $x - y = d$, we get $y = x - d$.
Therefore: $x + y = x + (x - d) = 2x - d < 0$
Since $x + y < 0$, we have $|x + y| = -(x + y) = -(2x - d) = d - 2x$.
So LHS $= d - 2x$ and RHS $= d$.
Since $x > 0$, we have $2x > 0$, which means LHS $= d - 2x < d =$ RHS.

Conclusion:
Cases I and II exhaustively cover all possible scenarios. In both cases, we have shown that LHS $<$ RHS, which means $|x + y| < |x - y|$.
Therefore, we have proven the stronger result: $\boxed{|x + y| \leq |x - y| \text{ for } x > 0, y < 0}$
Hence Proved. $\square$

Question 2

Problem Statement: Suppose you have an iterative algorithm for finding a fixed point of a function $F(x)$ on a certain interval. Write a short text (2-4 paragraphs) explaining the following:

1. How do you decide when to stop iterating?

2. What is your strategy for making sure that the computed solution is close to the exact solution?

DO NOT use the standard methodology and estimates from the textbook. Try to find other ways of accomplishing the goals above. Discuss briefly the advantages and disadvantages of your error control approach.

Solution:

Question 3

Problem Statement: Do Problem 23 c) from Section 3.4.

Problem 3.4.23 c): Find the order of convergence of the sequence:

$$x_n = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}}$$

Solution:

Question 4

Problem Statement: Do Problem 32 from Section 3.4.

Problem 3.4.32: Let $\frac{1}{2} \leq q \leq 1$, and define $F(x) = 2x - qx^2$. On what interval can it be guaranteed that the method of iteration using F will converge to a fixed point? (This problem is related to Problem 3.2.5, p. 90.)

Related Problem 3.2.5: What is the purpose of the following iteration formula?

$$x_{n+1} = 2x_n - x_n^2 y$$

Identify it as the Newton iteration for a certain function.

Solution:

Analysis of Related Problem 3.2.5

First, let's identify what the iteration formula $x_{n+1} = 2x_n - x_n^2 y$ represents as a Newton iteration.

The Newton-Raphson method for solving $f(x) = 0$ has the form:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Comparing with our given iteration:

$$x_{n+1} = 2x_n - yx_n^2 = x_n - (-x_n + yx_n^2)$$

Therefore:

$$\frac{f(x_n)}{f'(x_n)} = yx_n^2 - x_n$$

To solve for $f(x_n)$ we'll have to use integration. Let's temporarily switch to x as a variable instead of x_n.

$$\frac{f(x)}{f'(x)} = yx^2 - x$$

or,

$$\frac{f(x)}{\frac{df(x)}{dx}} = yx^2 - x$$

Cross-multiplying with $\frac{1}{dx}$ on both sides:

$$\frac{f(x)}{df(x)} = \frac{yx^2 - x}{dx}$$

Equivalently, we can write:

$$\frac{df(x)}{f(x)} = \frac{dx}{yx^2 - x}$$

Integrating both sides:

$$\ln|f(x)| = \int \frac{dx}{yx^2 - x} = \int \frac{dx}{x(yx-1)}$$

Dividing numerator and denominator or RHS by y:

$$\ln|f(x)| = \int \frac{1/y}{x(x-1/y)} dx$$

Therefore:

$$\ln|f(x)| = \int \left(\frac{1}{x - 1/y} - \frac{1}{x} \right) dx$$

$$\text{or,}\;\; \ln|f(x)| = \ln\left|x - \frac{1}{y}\right| - \ln|x| + \ln k \;,\; k \neq 0$$

$$\text{or,}\;\; \ln|f(x)| = \ln\left|\frac{k(x-\frac{1}{y})}{x}\right|$$

Taking the exponential:

$$f(x) = \frac{k}{x}\left(x - \frac{1}{y}\right)$$

Giving us the root of $f(x)$ or alternatively the fixed point of $F(x)$ at

$$x = r = \frac{1}{y}$$

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